Evaluation and Suppression of Vibrational Noise for Cryogenic Surface Electrode Ion Traps

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Cryogenic ion traps offer substantial advantages over their room-temperature counterparts, particularly in terms of stable ion confinement and cooling. However, this technological advancement has introduced a significant challenge, namely the generation of additional vibrational noise. This noise is pronounced during the compression and expansion phases of the Gifford–McMahon cycle refrigerator and stems from the mechanical operation of the cold head. The residual vibrational noise can modulate the Rabi oscillation frequency, thereby compromising the fidelity of the quantum logic gate. In this study, we quantitatively assess the impact of residual vibrational noise and introduce an innovative strategy for its mitigation. This strategy was designed to reduce superimposed potential fluctuations within a cryogenic surface electrode ion trap. The measured residual vibrational noise was used for real-time feedback control. Addressing the inverse determination of the compensating potential for a stable superimposed trapping potential is an inherently ill-posed problem considering known measurement noise results. By applying quadratic programming, we numerically calculated the optimal time series for the compensation potential. An ensemble of these compensation voltages was applied in real time to the static compensation electrodes. Although this method is in the theoretical stage, we believe that it can effectively suppress the translation induced by vibrational noise and foster the creation of a stable superimposed harmonic potential. The deployment of this technique is expected to substantially improve the accuracy of quantum operations, thereby fueling the evolution of quantum technologies.

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1. Introduction. Trapped ions have been used as platforms for frequency standards,[1–3] precision metrology,[4] and quantum information processing (QIP).[5,6] Recently, quantum computers have been experimentally demonstrated as superior to conventional supercomputers.[7–9] Trapped-ion quantum computing (TIQC) is a potential platform for constructing scalable quantum computers with high fidelity for elementary quantum operations and large overall quantum volume.[10,11]

TIQC was proposed by Cirac and Zoller.[12] In a linear Paul trap, the ions are trapped by an RF field in the transverse direction and a static electric potential in the axial direction. To prepare and manipulate the initial state, the ions were cooled to near the motional ground state using laser cooling techniques. Ion qubits are encoded in the ion hyperfine levels or Zeeman split levels, such as optical qubits[13,14] and microwave qubits.[15,16] Quantum logic gates can be implemented via different pulsed laser interactions with the encoded ions. Currently, single-qubit quantum logic gates achieve a fidelity of 99.9999%.[17] The fidelity of the two-qubit quantum gate has reached 99.9%.[18]

To scale up the qubits, Wineland[19] proposed a quantum charge-coupled device (QCCD) architecture. In this architecture, the ion trap is divided into different regions for different functions, such as loading, storage, and manipulation. By performing ion-transport operations, ions from multiple regions are entangled to extend the qubits. Therefore, a surface electrode ion trap (SEIT) was designed to transport trapped ions for quantum computation.[20–25] The electrodes in SEIT are mounted on a single plane,[26] which is distinct from the multifaceted 3D structural characteristics of blade Paul traps. They can be constructed using photolithography with a small size and complex configurations. Recently, a trapped-ion quantum computer (QCCD) was developed by Quantinuum Corporation. As presented in Ref. [27], the team successfully realized 32-ion single-qubit quantum logic gates with an average fidelity of 99.97% and fully connected two-qubit quantum logic gates with a fidelity of 99.8%.

However, SEIT faces several challenges. As the elec-

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trode size decreases, the trapped ions are only tens to hundreds of micrometers away from the electrode. The proximity of the ions to the electrodes results in a pronounced abnormal heating rate,[28,29] as observed in the experiments. Many experiments have shown that the heating rate significantly increases as the ion distance from the electrode surface decreases.[30–32] This heating rate is inherently linked to the spectral density of the electric field noise, which is characterized by the characteristic distance d from the ion to the electrode, the secular frequency ω, and the temperature T.[33,34] Extensive experimental data demonstrated an inversely proportional relationship between the heating rate and the fourth power of the characteristic distance.[35–39]

Over the last two decades, numerous experiments have shown that cryogenic ion trap systems can significantly reduce the heating rates associated with electric fields and Johnson noise.[35,39] Furthermore, they facilitate the creation of extremely high-vacuum environments, which in turn minimize collisions between trapped ions and background gas molecules. These improvements extend the lifetime and coherence time of the trapped ions.[31,40–42] For the experimental generation of a 4 K cryogenic condition in the sample chamber of the SEIT system, many research teams have utilized the Gifford–McMahon cycle refrigerator (GMCR).[40–44] However, owing to its operational mechanism, the cold head of a GMCR creates low-frequency vibrations during each cycle.[43,44] Residual vibrational noise can modulate the Rabi oscillation frequency, thereby adversely affecting the fidelity of quantum logic gates. This vibrational noise is highly detrimental to the quantum coherence manipulation.

In this study, by leveraging previously ascertained vibrational displacements using a homodyne quadrature laser interferometer (HQLI)[45] and employing numerical simulation techniques, we assessed the impact of vibrational noise scales on the coherent evolution and fidelity of a cryogenic surface electrode ion trap (CSEIT). In addition, we developed active vibration reduction techniques. In our method, we employed a dynamic compensation method for the potentials of multiple electrodes to stabilize the potential well of a confined ion. The ensemble of compensatory voltages derived from our numerical calculations was fed in real-time to the static compensation electrodes. By employing this approach, the deleterious effects of the trapped potential minimum vibration can be effectively mitigated, thereby enabling the efficient construction of a stable harmonic potential well. This methodology is expected to significantly bolster the stability and fidelity of the CSEIT, paving the way for enhanced performance in quantum information processing applications.

The remainder of this paper is organized as follows: Section 2 describes the apparatus used in the CSEIT and the measured frequency spectrum of the vibrational displacement. In Section 3, we introduce the quantum-dynamic theoretical model of the shaken time-dependent Hamiltonian and evaluate the effect of vibrational noise on the coherent state. In Section 4, we present an active vibration-reduction approach and calculate the inverse solution to the ill-posed problem for a compensated voltage set of resistive vibrational noise. Finally, we discuss and conclude this paper in Section 5.

2. Cryogenic Surface Electrode Ion Trap and its Measured Residual Vibrational Noise Spectrum. Owing to the GMCR cold head operating principle, vibrational noise was quantified at a displacement of 100 microns.[46] Passive vibration reduction techniques is essential in the CSEIT vacuum chamber, as shown in Fig. 1(a). Vibration mitigation involves flexible mounts and connections for the cold head, noncontact cooling via helium and a cold finger, and flexible metal bellows linking the sample and helium chambers to reduce vibrations from the compressor. In addition, we implemented strategies, such as buffer gases, flexible copper braids, and cold-finger heat exchange. All strategies are crucial for obtaining vibrational noise below 40 nm peak-to-peak.[47] The chamber included a 45 L/s gamma vacuum ion pump, pride cryogenic GMCR with 1.5 W at 4 K, and a sample vacuum chamber with our SEIT.

Our linear Paul SEIT device has a "five-wire trap" configuration. A more detailed description of SEIT was introduced in Refs. [48,49]. The SEIT consisted of 15 pairs of DC segments, as shown in Fig. 1(b). The DC electrodes were named 1a(b) to 15a(b) and used for axial confinement. The other electrodes, RF1(2) and GND, were used for transverse confinement and loaded with RF. According to experimental experience with a normal-temperature ion trap, radial confinement is provided by an RF potential of $V_0 = 150\,\mathrm{V}$ (0-peak) and frequency $\varOmega_{\mathrm{RF}} = 2\pi\times 22.657\,\mathrm{MHz}$. The axial confining potential was provided by DC voltages ($-10\,\mathrm{V}$ to $10\,\mathrm{V}$), which were applied to 15 pairs of DC segments. To suppress the residual vibrational noise, SEIT is processed in modular partitions according to different functions. As shown in Fig. 1(c), the DC electrodes named 1a(b) to 3a(b) and 13a(b) to 15a(b) are used to resist harmonic shaking by residual vibrational noise in the two blue bilateral dashed boxes. Therefore, these DC electrodes are referred to as compensatory DC electrodes. DC electrodes 4a(b) to 12a(b) are used to trap and manipulate the middle boxes with red solid lines, and are named the trap and manipulation DC electrodes. In a previous study, we measured the residual vibrational noise spectra and vibrational displacements, as shown in Figs. 1(d) and 1(e).[45] Several discrete low-frequency vibrational frequencies were dominant. This conclusion was confirmed in Ref. [47], which characterized the vibrational noise and the measurement method with a Michelson interferometer that is locked to a fringe by proportional-integral-derivative (PID) feedback control with a piezoelectric mounted mirror (PZT). After we created some vibration-reduction optimization designs for the experimental setup, the vibration amplitude was within 100 nm, even when the GMCR was on.

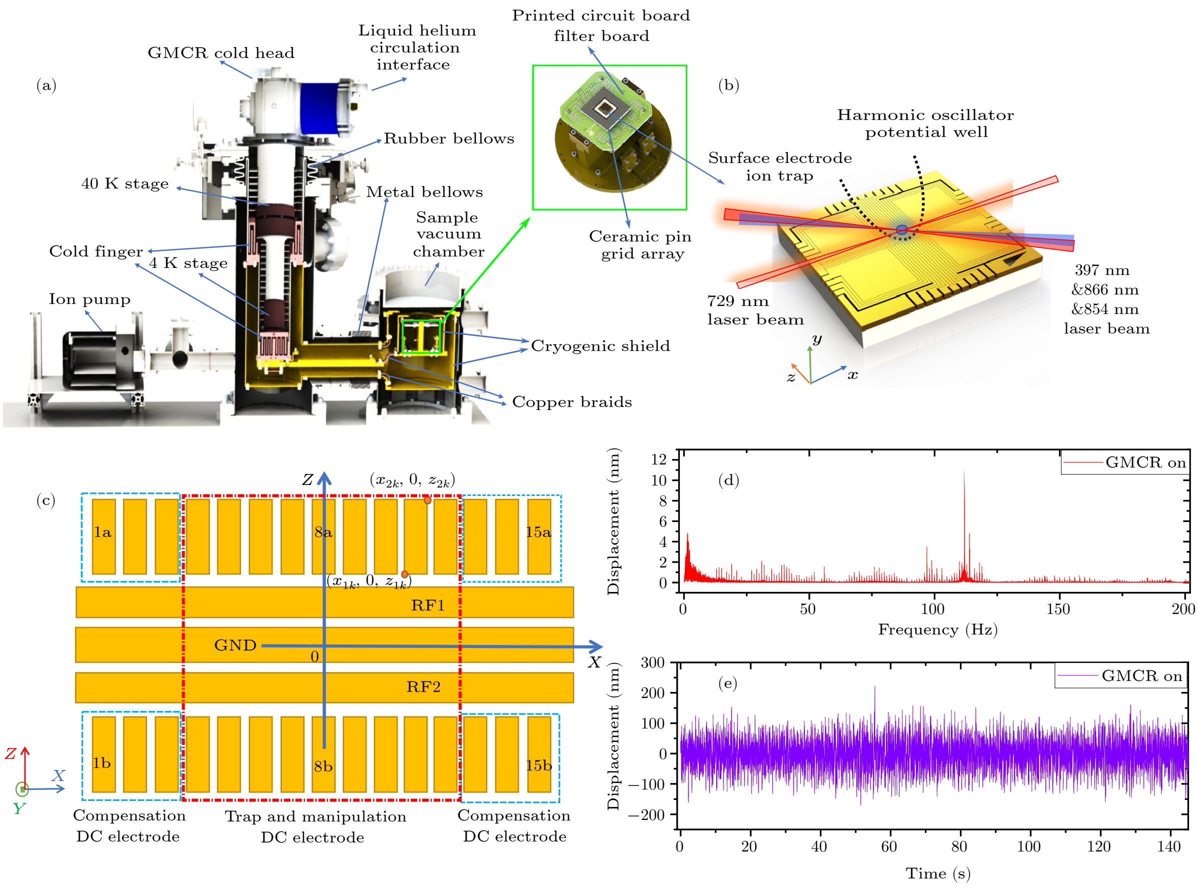

Fig. 1. Schematic diagram of the CSEIT and its measured residual vibrational noise spectrum. (a) Cutaway drawing of the CSEIT system. The bright green box shows the low-pass filter printed circuit board (PCB), ceramic pin grid array (CPGA) and SEIT. (b) Single Ca^+ ion trapped above the linear Paul SEIT and Doppler cooled by the 397 nm, 866 nm and 854 nm laser light. The quantum state is manipulated by a 729 nm laser along the axial direction. (c) Electrodes of a linear Paul SEIT with a "five-wire trap" configuration. The DC electrodes are named 1a(b) to 15a(b) above (down) for the radio-frequency electrodes RF1(2) and GND. The DC electrode from 4a(b) to 12a(b) is used for trapping and manipulation. The compensation DC electrodes 1a(b) to 3a(b) and 13a(b) to 15a(b) are used to resist harmonic shaking from residual vibrational noise. (d) Frequency spectrum of measured residual vibrational noise. (e) Displacement of measured residual vibrational noise.

3. Evaluation of the Effect of Vibrational Noise via Characteristic Low-Frequency Noise

3.1. Dynamics of a Time-Dependent Hamiltonian With a Shaken Harmonic Potential. Consider a single ion confined within a linear Paul trap. For simplicity, we focus solely on the dynamics of a harmonic oscillator in the axial direction, which is subjected to external vibrational noise in a slowly shaking potential well within a cryogenic ion trap system. In this ansatz, the Hamiltonian evolves in a time-dependent form, and is expressed as follows:

$$H = \frac{1}{2m}P^2 + \frac{1}{2}m\omega_{\rm axial}^2(X - X_0(t))^2, \tag{1}$$

where P and X are the momentum and position operators, respectively; $\omega_{\rm axial}$ is the axial trap frequency; m is the ion mass; and $X_0(t)$ is the position of the potential minimum. If $X_0(t)$ is controllable, ion transportation can be realized by manipulating the time-dependent transport function $X_0(t)$ for SEIT.[24,50–53] In the CSEIT system, $X_0(t)$ is shaken by external mechanical vibrations and the time-dependent potential $V(t) = \frac{1}{2}m\omega_{\rm axial}^2(X - X_0(t))^2$ is the shaken harmonic potential. As a time-dependent Hamiltonian problem, the dynamics of such systems can be solved directly via numerical integration of the Schrödinger equation

$$H(t)|\Psi(t)\rangle = i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle, \tag{2}$$

where $\Psi(t)$ is the wavefunction solution of the time-dependent Hamiltonian $H(t)$. The potential energy $V(t)$ satisfies the periodic boundary condition $V(t) = V(t+T)$, where T denotes the period. According to the Floquet theorem, a solution exists that is a superposition of Floquet states.

Therefore, we consider a general solution for evolution over any period. When vibrational noise exists, i.e., $X_0(t) \neq 0$, we transform the wave function $\Psi(t)$ to the remaining frame of the potential and define a state $|\chi(t)\rangle = D^\dagger(\tilde{X}_0(t))|\Psi(t)\rangle$, where D is a displacement operator $D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a)$, where $\tilde{X}_0(t) = X_0(t)/(2x_0)$ is the normalized displacement with $x_0 = \sqrt{\hbar/(2m\omega_{\rm axial})}$.

The state satisfies the Schrödinger equation:

$$i\hbar\partial_t|\chi(t)\rangle = (H_0 - \dot{X}_0(t)P)|\chi(t)\rangle, \tag{3}$$

where $H_0 = \frac{1}{2m}P^2 + \frac{1}{2}m\omega_{\rm axial}^2X^2$, P is the momentum operator whose definition is $P = -i\sqrt{\frac{\hbar m\omega_{\rm axial}}{2}}(a - a^\dagger)$, where $a^\dagger$ and a are the raising and lowering operators of H_0. Referring to Ref. [51], we review the derivation as follows. In the interaction picture, state becomes

$$|\chi^I(t)\rangle = U_0^\dagger(t)|\chi(t)\rangle, \tag{4}$$

where $U_0(t) = \exp(-iH_0t/\hbar) = \exp(-i\omega_{\rm axial}(a^\dagger a + \frac{1}{2}))$ is the time evolution operator. Therefore, the interaction Hamiltonian satisfies $i\hbar\frac{\partial}{\partial_t}|\chi^I(t)\rangle = H_{\rm int}(t)|\chi^I(t)\rangle$. Combining Eq. (3), we obtain the interaction Hamiltonian as

$$\begin{aligned} H_{\rm int}(t) &= U_0^\dagger(t)(-\dot{\tilde{X}}_0(t)P)U_0(t) \\ &= -\dot{\tilde{X}}_0(t)(a^\dagger e^{i\omega_{\rm axial}t} - ae^{-i\omega_{\rm axial}t}) \\ &= \dot{\tilde{u}}(t)a^\dagger - \dot{\tilde{u}}^*(t)a, \end{aligned} \tag{5}$$

where $\tilde{u}(t)$ is defined as $\tilde{u}(t) = -\int_0^t \dot{\tilde{X}}_0(\tau)e^{i\omega_{\rm axial}\tau}d\tau$, The time evolution can be calculated via the Magnus expansion as

$$\begin{aligned} \tilde{U}(t) = \exp\bigg\{ &\int_0^t H_{\rm int}(t_1)dt_1 \\ &+ \frac{1}{2}\int_0^t dt_1 \int_0^{t_1} [H_{\rm int}(t_1), H_{\rm int}(t_2)]dt_2 \bigg\} \\ = D(\tilde{u}(t))&e^{i\varphi(t)}. \end{aligned} \tag{6}$$

In the above equation, $\varphi(t)$ is an overall phase of the ion defined as $\varphi(t) = \frac{1}{2}\int_0^t[\tilde{u}(t')\dot{\tilde{u}}^*(t') - \tilde{u}^*(t')\dot{\tilde{u}}(t')]dt'$. Because the commutators $[H_{\rm int}(t_1), H_{\rm int}(t_2)]$ are scalar, the higher-order terms in the Magnus expansion vanish. By combining the above equations, the wave function of the trapped ion is denoted by

$$|\Psi(t)\rangle = D(\tilde{X}_0(t))e^{-iH_0t/\hbar}D(\tilde{u}(t))D^\dagger(\tilde{X}_0(0))e^{i\varphi(t)}|\Psi(0)\rangle. \tag{7}$$

Based on the aforementioned derivation, provided that the measured vibrational displacement $X_0(t)$ is ascertained, the Hamiltonian of the harmonic potential and the associated time-dependent evolution operator can be numerically determined. Consequently, the temporal evolution of the wave function was derived. The wave function $|\Psi(t)\rangle$ is closely related to the initial state $|\Psi(0)\rangle$ and vibrational displacement $X_0(t)$ at any time t. If the initial state $|\Psi(0)\rangle$ is the ground state of the harmonic oscillator minimal position at $\tilde{X}_0(0) = 0$, then the motion state remains coherent during the entire process, with $|\Psi(t)\rangle = |\alpha(t)\rangle$.

3.2. Evaluation of the Effect of Residual Vibrational Noise on The Shaken Harmonic Potential. According to the experimental measurements of the residual vibrational noise by several research groups,[44,47,54] the following reasonable assumptions can be made for such low-frequency discrete signals: This vibrational noise has a certain periodicity, and when random noise with small vibrational amplitudes and errors owing to measurement accuracy is neglected, only a few basis functions can be superimposed to properly fit the vibrational noise signal. These assumptions provide a theoretical basis for evaluating vibrational noise via numerical simulations.

To approximate the experimental parameters, we considered four sets of conditions at different residual vibrational noise levels ranging from 10 nm to 110 nm. To evaluate the effect of vibration, we simulated the relationship between the phonon number distribution and vibrational noise. The details of these four sets are presented in Table 1. Four groups of different noise levels were considered, typically based on the vibrational noise spectrum captured by the experimental HQLI, as shown in Fig. 1(d). Furthermore, we considered the experimental results of Ref. [47]. Other parameters can be set to reflect the influence of the noise level on the excitation of the phonon state population and fidelity of the coherent state. There may be negligible errors in the selection of the four sets of simulation parameters, which were based on the factors described above.

A vibratory noise signal can be expressed in a discrete Fourier series form $X_0(t) = \sum_{\omega_i=-\infty}^{\infty} F[\omega_i]e^{i\omega_i t}$, where $F[\omega_i]$ represents the complex amplitude. A real-valued function of vibrational noise can be approximately expressed as

$$X_0(t) \approx \sum_{i=0}^{N=3} A(\omega_i)\sin(2\pi\omega_i t), \tag{8}$$

where ω_i denotes the frequency corresponding to the larger amplitude in the spectrum. It can be approximated as a superposition of cosine functions and the same assumption can be made by starting with different initial phases or translations. Here, we first consider the effect of vibrational noise strength on fidelity, neglecting the effect of the initial phase of the vibrational displacement signal in the numerical simulations. The initial phase of the sinusoidal basis function for several superpositions of vibrational displacements was assumed to be zero. We assumed only the effects of different residual noise amplitude scales on the fidelity of the prepared initial phonon state and presented the simulation results via analytical calculations, as shown in Subsection 3.1, and the QuTiP numerical simulation toolkit.[55,56] Using the methods described above, the

Table 1. Numerical simulation the effect of residual vibrational noise with different amplitude parameters. Here, we demonstrate the approximation of residual vibrational noise at different levels by considering only four ensemble parameters with three characteristic frequencies, which are based on the vibrational noise spectrum.

Groups of different noise levels	$A\,(\omega_1 = 2\pi\cdot 40\,\mathrm{Hz})$	$A\,(\omega_2 = 2\pi\cdot 100\,\mathrm{Hz})$	$A\,(\omega_3 = 2\pi\cdot 120\,\mathrm{Hz})$	Maximum amplitude
Case I	1.70 nm	2.00 nm	7.00 nm	9.83 nm
Case Ⅱ	1.7 nm	4.0 nm	15.0 nm	19.76 nm
Case Ⅲ	10.7 nm	20.0 nm	35.0 nm	60.23 nm
Case Ⅳ	20.7 nm	40.0 nm	65.0 nm	109.73 nm

phonon number $\langle \hat{n} \rangle$ and occupation probability $P(i)$ of the coherence are calculated as follows:

$$\begin{cases} \langle \hat{n} \rangle = \mathrm{Tr}\,[\rho a^\dagger a], \\ P(i) = \langle i|\rho|i\rangle, \quad i = 0, 1, 2, 3, \ldots, \end{cases} \tag{9}$$

where $|i\rangle$ is the Fock number state (phonon state), $\rho = |\Psi(t)\rangle\langle\Psi(t)|$ is the density operator, and $|\Psi(t)\rangle$ is expressed by Eq. (7). Here, we assume that $|\Psi(0)\rangle = |0\rangle$, and that the initial state of an ion trapped in a harmonic potential trap is in the coherent ground state after laser cooling. The fidelity of the arbitrary state ρ and the initial pure states $|\Psi(0)\rangle$ were obtained according to Ref. [57] as follows:

$$F(|\Psi(0)\rangle, \rho) = \mathrm{Tr}\left[\sqrt{\langle\Psi(0)|\rho|\Psi(0)\rangle}\right]. \tag{10}$$

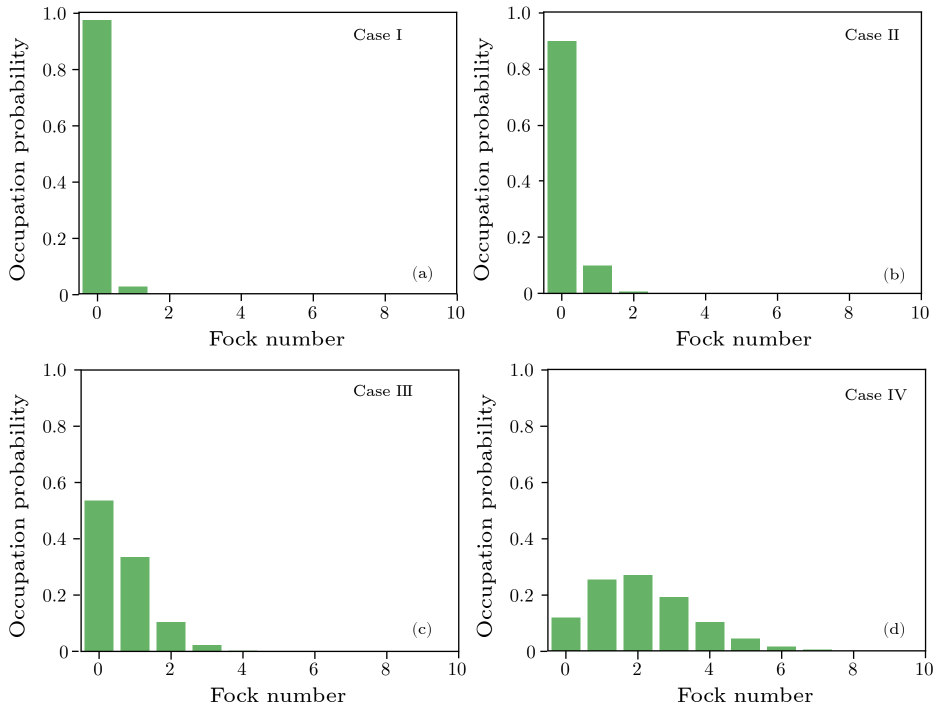

Fig. 2. Numerical simulation of the occupation probability with different noise levels when the vibrational noise is at the maximum level. (a)–(d) correspond to the occupation probability of the Fock state at maximum displacement with four different noise magnitude parameters for Case I, Case Ⅱ, Case Ⅲ, and Case Ⅳ, respectively. The above four figures correspond to the time-dependent change in the Hamiltonian owing to vibrational noise, which causes the coherent state to be excited from the initial ground state to the high phonon state with a certain probability. $\omega_{\mathrm{axial}} = 2\pi \cdot 250\,\mathrm{kHz}$.

The results of the occupation probability $P(i)$ are shown in Fig. 2. With increasing vibrational amplitude, the excitation probability of the phonon state increased. As the noise strength increases, there is a certain probability of the phonon states being excited; thus, the average number of phonons in the system increases. Therefore, the shaking-trapping potential increases the average phonon number of the system, which causes the system to thermalize. In Cases I and Ⅱ, the probability that the phonon state is occupied is almost equal for the $|0\rangle$ and $|1\rangle$ states for vibrational displacements smaller than 20 nm. This conclusion is confirmed in the present study. When the vibrational amplitude was less than 20 nm, the experimental results indicated that the mean phonon number of the two-bit bright state in the entangled Ising model followed an exponentially damped sinusoidal function. At this vibrational level, the excitation of the average phonon number was not induced. Considering the parameters of Cases Ⅲ and Ⅳ, when the vibrational displacement is larger than 20 nm there is a certain probability that the phonon state population number will be excited to a higher-dimensional state greater than or equal to $|2\rangle$ states. The results of the numerical simulations indicate that when the potential well is shaken beyond a certain critical value, a significant heating effect is induced, and the average phonon number of the ions increases sharply. The ground state of a general harmonic oscillator is a Gaussian wave packet with a full width at half maximum (FWHM) of approximately 10 nm for the ground-state wave function. At this point, if an external vibration causes the harmonic potential well to shake and the trapped ions in the potential well to move, the amplitude of the motion increases and the energy of the system is equivalent to an increase. Shaking of the external potential well was not evident and the Gaussian wave packet function was insufficient to distinguish and reflect the heating effect. The effect of external vibrational heating is highlighted when the vibrational noise level during a period exceeds a certain critical value where the relative shift in the Gaussian wave packet function during the vibrational period is sufficient to distinguish between the two wave-packet functions.

We numerically simulated fidelity versus vibrational displacement for the parameters in Cases I, Ⅱ, Ⅲ, and Ⅳ. The fidelity decreases when the amplitude of the vibrational noise increases. However, there is a certain phase delay between fidelity and displacement, and the overall relationship is positive, as shown in Fig. 3. Considering the parameters in Case I, when the vibration displacement was less than 10 nm, the maximum fidelity error was less than 2%, as shown in Fig. 3(a). When the vibration was below 20 nm, the fidelity exceeded 92% as shown in Fig. 3(b). Considering the parameters of Case Ⅲ, when the vibration displacement is approximately 60 nm, the probability of the maximum fidelity error is greater than 50%, as shown in Figs. 3(c) and 3(d). When the displacement was small, the heating of the system was also small, and the fidelity was less affected.

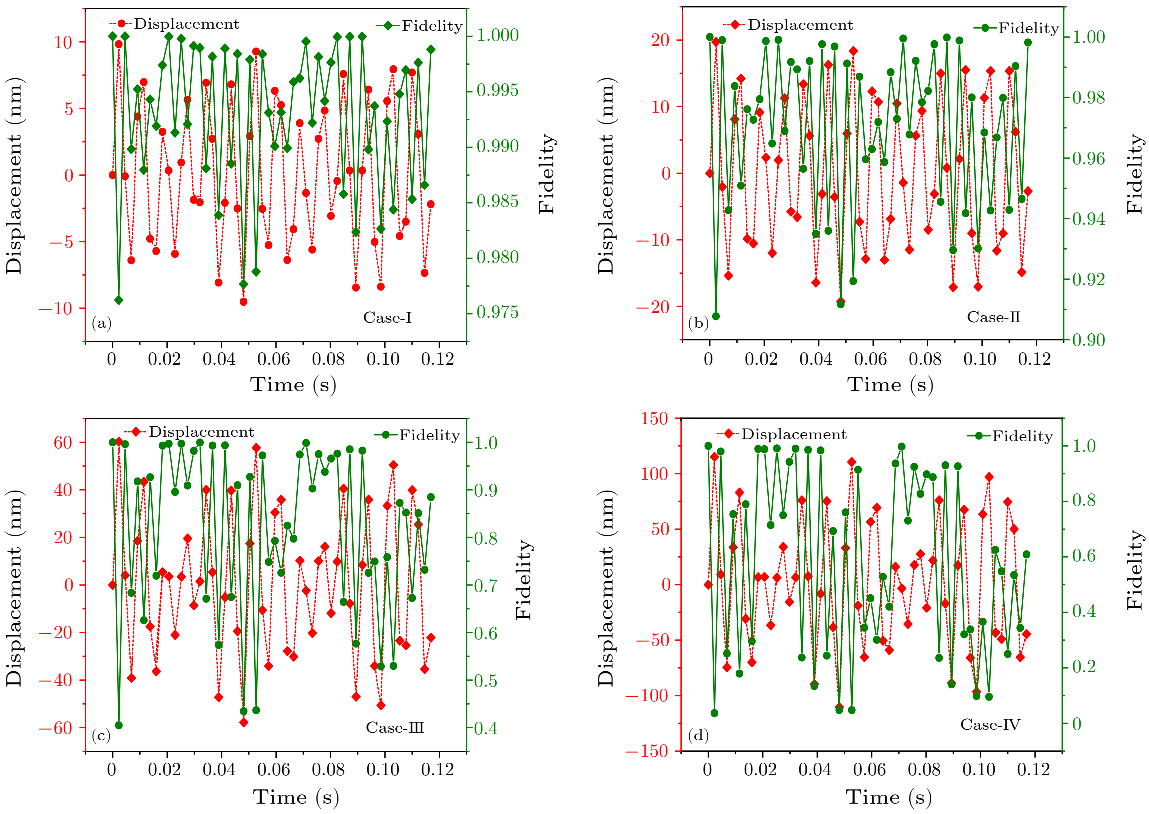

Fig. 3. Numerical simulation of the fidelity with different amplitude noise. The green solid line represents the fidelity between the initial state and the time-dependent evolution of the coherent state, and the red dashed line represents the displacement of the potential well owing to vibrations. (a)–(d) correspond to the results of numerical simulations with parameters according to Case I, Case Ⅱ, Case Ⅲ, and Case Ⅳ, respectively. $\omega_{\rm axial} = 2\pi \cdot 250\,\rm kHz$.

4. Suppression of Vibrational Noise

4.1. Scheme for Resisting Vibrational Noise by Suppressing Time-Dependent Potential Shaking. According to the numerical simulation results in Subsection 3.2, when the displacement noise is substantial, it results in significant heating of the system and severely affects fidelity. Therefore, it is imperative to devise methods to mitigate the substantial vibrational noise. Consequently, we introduce our theoretical framework aimed at suppressing vibrational noise by leveraging the characteristics of multiple electrode pairs present in the CSEIT. This approach was designed to ensure the stability of the potential well and minimize perturbations induced by vibrational noise in the system dynamics. By applying an ensemble of these computed compensated voltages to static electrodes in real time, we effectively suppressed the translational effects caused by vibrational noise, thus facilitating the establishment of a stable harmonic potential well.

Before investigating vibration suppression techniques for cryogenic ion trap systems, it is essential to recognize that the frequency spectrum of the vibrational noise is characterized by discrete low-frequency components. Furthermore, it is reasonable to hypothesize that these vibrations are typically induced by the GMCR and its associated compressor, which resonate at specific frequencies and exhibit higher harmonics. The characteristics of these vibrations and the displacement amplitude during the vibration process can be measured precisely using a Michelson interferometer.[47] By employing a PID feedback control system, the measurement range of the vibration amplitude can be extended to more than half the wavelength. Consequently, real-time measurements of the vibration displacement of the system could be performed.

Although various vibration-isolation techniques have been used, some residual acoustic vibration noise remains. Considering the characterization of vibrational noise, we propose a scheme that resists vibrational noise by suppressing the time-dependent potential shaking. The framework and flowchart are presented in Fig. 4. In our framework, we first measured vibrational noise using HQLI.[45] The measurement results of the vibration are used to solve an ill-posed problem in which the compensating voltage set can maintain the trap center under vibration conditions. A quadratic programming optimization method was used to calculate the compensated voltage sequence. Finally, the compensation voltage sequence was applied to the compensation DC electrodes using a high-speed arbitrary wave generator (AWG) in real time.

The ideal harmonic oscillator potential without vibrational noise in a cryogenic ion trap system can be expressed as $\Phi_{\rm ideal} = (1/2)\beta X^2$. The real harmonic oscillator potential with the residual vibration noise is defined as $\Phi_{\rm real} = (1/2)\beta(X - X_0(t))^2$. The time-dependent compensation potential can be computed using the following equation:

$$\Phi_{\rm compensation} = \Phi_{\rm ideal} - \Phi_{\rm real}. \tag{11}$$

The compensated confinement potential was calculated using the real-time vibrational displacement of HQLI. The compensation principle is illustrated in Fig. 5. The larger the displacement of the oscillator, the larger is the slope of the compensating potential, which implies a larger magnitude of the compensating electrode potential. The resonant potential was robustly maintained in a real-time dynamic regime by loading a sequence of compensating potentials onto the compensating electrodes.

Because the trap potential of the SEIT is powered by a voltage source, the harmonic potential should be constructed by the superposition of all charged DC electrodes. To construct the harmonic trapping potential, the multipoles must be mapped onto the electrode voltages. A unit voltage potential ϕ_k was created when a voltage of 1 V was applied to the kth electrode and when 0 V was applied to all other electrodes. According to the superposition principle, the total potential of the SEIT is equal to the sum of independent potentials:

$$\Phi = \sum_k V_k \cdot \phi_k, (k = 1, \ldots, N), \tag{12}$$

where the voltage V_k is applied to the kth electrode. We consider the construction of the ideal harmonic oscillator potential and solve this ill-posed problem.

There are several methods for determining the unit voltage potential ϕ_k, including the finite difference method (FDM), finite element method (FEM), and boundary element method (BEM). The potential outcomes of the FDM and FEM are not smooth. The precision potential must be meshed using a nanoscale spatial grid, which involves prohibitive computations. The BEM provides high accuracy and smooth potential. For simplicity, we used the BEMSolver tool kit[58] and analytical model[59] calculations

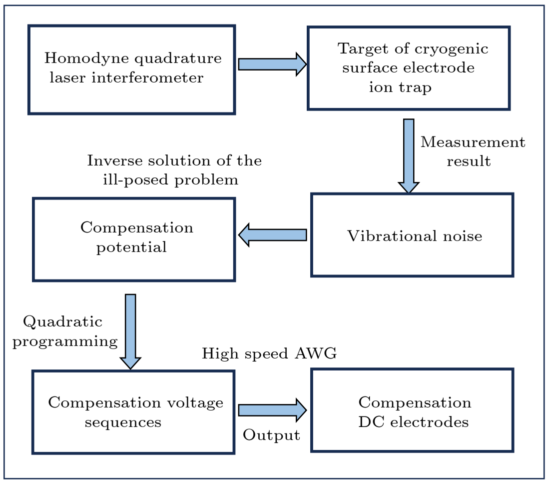

Fig. 4. Framework for suppressing vibrational noise.

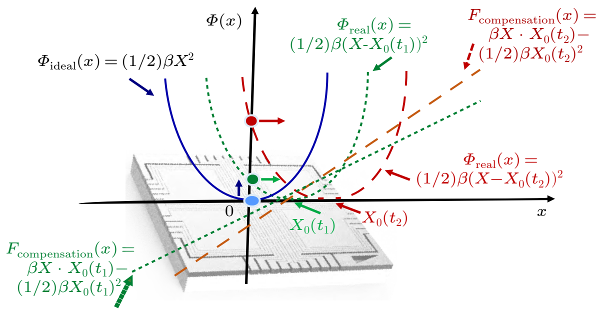

Fig. 5. Time-dependent harmonic oscillator potential. The blue solid line is the ideal harmonic potential $\Phi_{\rm ideal}$ when $t = 0$ at the beginning. The green quadratic function and linear dotted line are the real harmonic potential $\Phi_{\rm real}$ and compensation potential $\Phi_{\rm compensation}$ at time $t = t_1$, respectively; subsequently, the position of the harmonic potential minimum from point O becomes $X_0(t_1)$. The red quadratic function and linear dashed line are the real harmonic potential $\Phi_{\rm real}$ and compensation potential $\Phi_{\rm compensation}$ at time $t = t_2$, respectively; the position of the harmonic potential minimum becomes $X_0(t_2)$.

of the electrostatic potential of a planar electrode. Referring to the analytical model of M. G. House's theory,[59] the static potential of the k^{th} rectangular electrode can be calculated as

$$\begin{aligned}
&\phi_k(x, y, z) \\
&= \frac{V}{2\pi}\Bigg\{\arctan\left[\frac{(x_{2k}-x)(z_{2k}-z)}{y\sqrt{y^2+(x_{2k}-x)^2+(z_{2k}-z)^2}}\right] \\
&\quad - \arctan\left[\frac{(x_{1k}-x)(z_{2k}-z)}{y\sqrt{y^2+(x_{1k}-x)^2+(z_{2k}-z)^2}}\right] \\
&\quad - \arctan\left[\frac{(x_{2k}-x)(z_{1k}-z)}{y\sqrt{y^2+(x_{2k}-x)^2+(z_{1k}-z)^2}}\right] \\
&\quad + \arctan\left[\frac{(x_{1k}-x)(z_{1k}-z)}{y\sqrt{y^2+(x_{1k}-x)^2+(z_{1k}-z)^2}}\right]\Bigg\},
\end{aligned} \tag{13}$$

where $(x_{1k}, 0, z_{1k})$ and $(x_{2k}, 0, z_{2k})$ are the positions of the opposite corners of the kth electrode, and V is the voltage applied to the electrode, with $V = 1\,\mathrm{V}$.

As aforementioned, the key to suppressing the oscillations of the harmonic potential well owing to vibrational noise is the calculation of the compensating electrode voltage that generates the compensating potential. We summarize the problem and describe it using simple mathematical terms. This problem can be transformed into a quadratic programming optimization problem. Therefore, the voltage sequence of the compensation electrode can be solved as follows:

$$\begin{cases} \mathrm{Min}_V \|V_i\phi_i - \varPhi_{\mathrm{compensation}}\|_2, \quad (i = 1, \ldots, N), \\ \sqrt{\frac{|e|\partial^2\varPhi_{\mathrm{ideal}}}{m\partial x^2}} = \omega_{\mathrm{axial}}, \\ lb \le V_j \le ub, \quad (j = 1, \ldots, N), \\ \varPhi_{lb} \le \phi_k \cdot V_k \le \varPhi_{ub}, \quad (k = 1, \ldots, N), \end{cases} \tag{14}$$

where lb and ub are the minimum and maximum voltages of the electrode, respectively, and $\varPhi_{lb}$ and $\varPhi_{ub}$ are the minimum and maximum spatial potentials, respectively. The minimum L^2-norm $\|V_i\phi_i - \varPhi_{\mathrm{compensation}}\|_2$ is an objective function. The optimal solution of Eq. (14) is the compensation voltage sequence for resisting vibrational noise. Although the compensated electrode voltage is solved using quadratic programming, it is possible that this is only a locally optimal solution and not a globally optimal solution. However, to compensate and mitigate the adverse effects of vibrational noise, it is necessary to achieve this goal.

4.2. Potential Sequence of the Compensation Electrode for Suppressing Residual Vibrational Noise. Considering the effect of vibrational noise on the experimental fidelity, the observed results denoted by the Monroe group indicate that the peak-to-peak values are within 40 nm and that the shaking effect can be neglected.[47] When the vibrational amplitude satisfies this condition, their experimental results show that the single-bit Rabi oscillation fits a sinusoidal function, and the mean phonon number of the two-bit bright state in the entangled Ising model fits an exponentially damped sinusoidal function. To reduce the influence on the modulation Rabi frequency of the laser and avoid affecting the fidelity, we maintained a stable

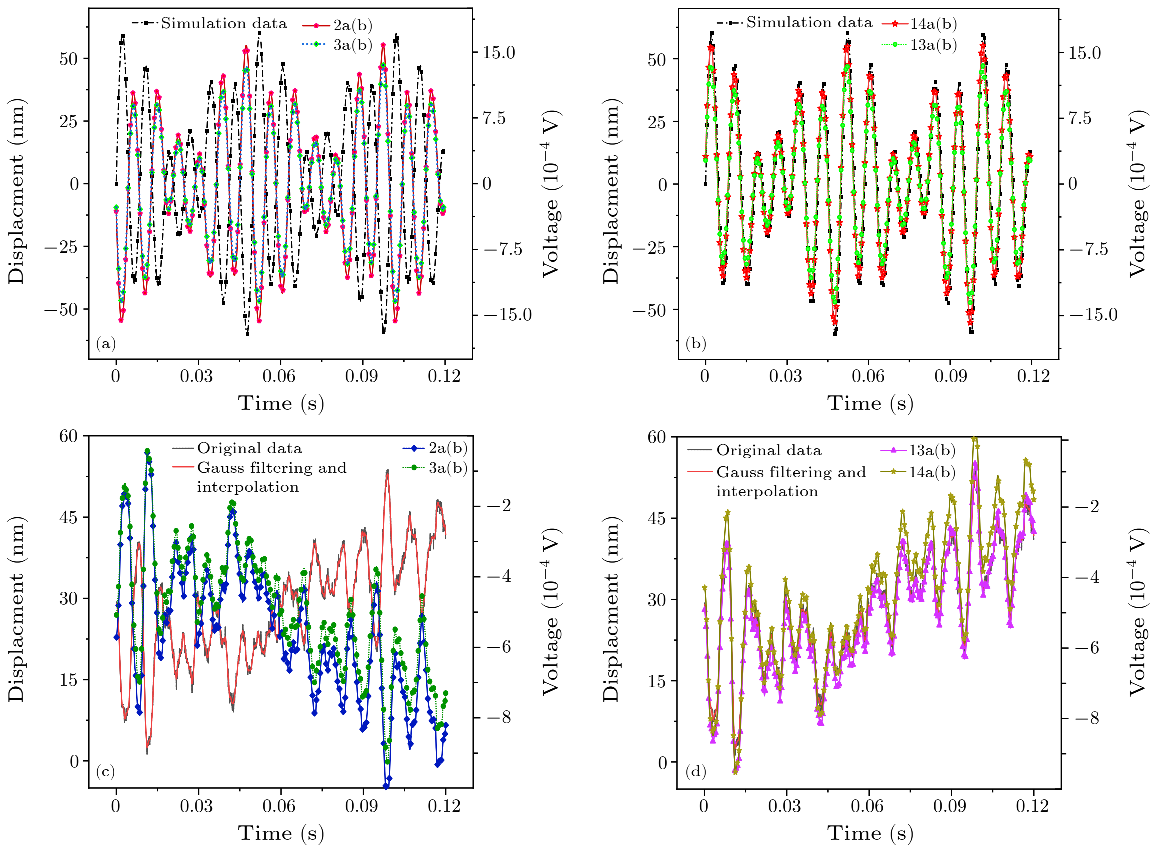

Fig. 6. Compensation voltages. (a) Black dashed-dotted line represents the residual vibrational noise considering Case Ⅲ parameters, red solid line represents the compensation voltage of compensation electrode 2a(b), and blue dotted line represents the compensation voltage of compensation electrode 3a(b). (b) Black dashed-dotted line similarly represents the residual vibrational noise considering Case Ⅲ parameters, the green dashed-dotted line represents the compensation voltage of compensation electrode 13a(b), and the red solid line represents the compensation voltage of compensation electrode 14a(b). Similarly, we selected the real vibrational data and calculated the compensated electrode voltages. The black dashed lines in (c) and (d) represent the measured discrete vibration displacement data, and the red solid lines represent the vibration displacement after Gaussian smooth filtering and interpolation. (c) The blue diamond mark and the green dot mark line represent the compensating electrode 2a(b) and 3a(b) voltages, respectively. (d) The purple triangle mark line and the earthen yellow five-pointed star mark line represent the 13a(b) and 14a(b) voltage of the compensating electrode, respectively.

phase between the harmonic oscillator and the interacting laser by PID feedback control of the piezo-controller mirror (PZT) to realize vibrational noise measurement and compensation. This method used a proportional-integral-derivative (PID) controller to lock the interferometric intensity signals of the measuring and reference arms in a Michelson interferometer to measure the vibrational displacement of the target. Similarly, compensation for vibrational noise can be realized.[60]

In our strategy, during the vibration of the CSEIT induced by the cryogenic system, the compensatory voltages is controlled in real time to maintain the confinement position of the trapping potential. This strategy effectively prevents the excitation of the ion's motional modes. For the case where the amplitude value of the vibrational amplitude exceeds 20 nm and considering multiple DC electrodes of the CSEIT, we propose stabilizing the position of the harmonic potential trap by adding a compensation potential in real time, which can resist the influence of low-frequency vibrational noise on a small scale.

The simulation results for the proposed solution, obtained using the least-squares method, are shown in Fig. 6, where the residual vibrational noise is provided by the Case Ⅲ parameters to be used to simulate and calculate compensation voltages. The potential calculation model used in this study is an analytical approximation model.[59] In Figs. 6(c) and 6(d), the black dashed line represents the discrete vibrational displacement data measured by the experiment, and the red solid line represents the vibrational displacement after Gaussian smooth filtering and interpolation. The compensated electrode voltages were calculated using the real vibrational data segments. The underlying computational model was calculated using the BEMSolver numerical tool.[58]

Our strategy was to select only 2a(b), 3a(b), 13a(b), and 14a(b) as the compensation electrodes. When the vibration position shifted in the positive x-axis direction, the compensation voltages of 2a(b) and 3a(b) were inversely related to the vibrational displacement, as shown in Figs. 6(a) and 6(c). The compensation potentials of 13a(b) and 14a(b) are linearly and positively proportional to the vibrational displacement, as shown in Figs. 6(b) and 6(d). The numerical results are consistent with this assumption.

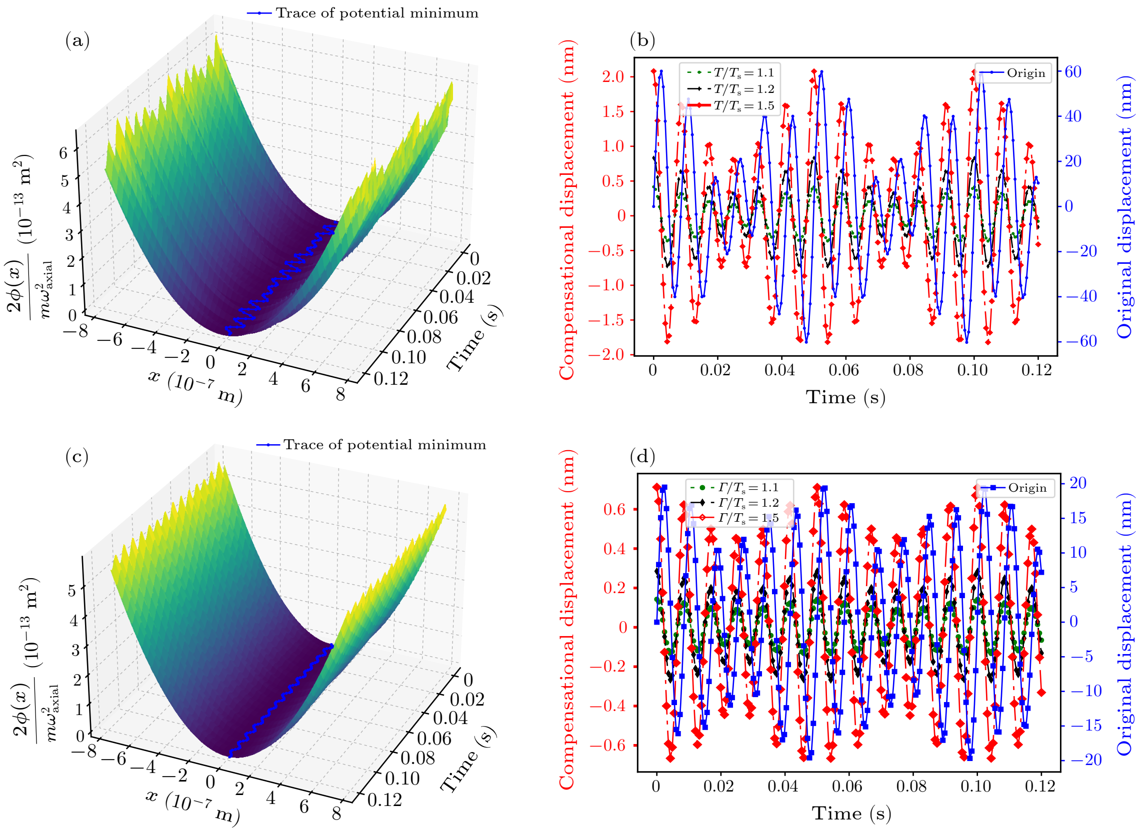

Fig. 7. Stabilization of the position of the potential minimum after application of the compensation voltages. (a) Trace of the potential minimum with residual vibrational noise. The blue solid line represents the trace of the potential minimum. (b) Vibrational level before and after compensation. The blue solid line represents the original displacement, and the green dotted line, black dashed line and red dashed-dotted line represent the compensated displacements considering $T/T_{\rm s} = 1.1$, $T/T_{\rm s} = 1.2$, and $T/T_{\rm s} = 1.5$, respectively. (a) and (b) under the condition corresponding to CASE-Ⅲ parameters, the effect of vibration displacement suppression by electrical methods with different feedback periods is numerically simulated. Similarly, (c) and (d) correspond to the results of numerical simulation suppresion vibrational noise under the condition of CASE-Ⅱ parameters.

In addition, we considered the effects of different compensated feedbacks and sampling bandwidths of the measured vibrational displacements. A measured displacement sampling period of $T_{\mathrm{s}} = 0.1\,\mathrm{ms}$. We considered the total compensation period as $T = T_{\mathrm{s}} + T_{\mathrm{cal}} + T_{\mathrm{del}}$, where T_{cal} is the calculated time of the compensated voltage and T_{del} is the delay time of loading the compensated voltage. Therefore, we obtained the simulated results of the trace of the potential minimum and the vibrational level before and after compensation, as shown in Figs. 7(a) and 7(b). Provided that the sampling bandwidth for measurement and the feedback control bandwidth for noise suppression are sufficiently high, the electrode voltage can ideally be compensated in real time. This ensured that the vibration of the minimum potential energy point within the composite trapping potential remained within a small amplitude of 10 nm. Consequently, a potential well was maintained to not adversely affect the coupling strength of the laser-ion interaction or the fidelity of the quantum logic gate.

To demonstrate the superiority of this electrical method in suppressing the vibration of the harmonic potential well caused by mechanical vibration, even under small vibration conditions, such as referring to CASE-Ⅱ parameters, the vibration magnitude is approximately 20 nm. We propose to use the least squares method to reverse the ill-posed problem and obtain the electrode voltage compensation, which can further alleviate the impact of mechanical vibrational noise. The numerical results are presented in Figs. 7(c) and 7(d).

5. Discussion and Conclusion. We conducted an exhaustive analysis of the vibrational effects in the CSEIT. Our numerical simulations, informed by the vibrational noise spectrum obtained from our experiments, are crucial for evaluating the fidelity of coherent states with varying vibrational noise amplitudes. Furthermore, in the absence of laser interaction, the oscillating potential of the trap induces the heating of the trapped ions, which in turn increases the phonon distribution. Although we numerically simulated the effect of the vibration-induced oscillation of the harmonic potential well on the fidelity of the external phonon states, this analysis is not sufficiently comprehensive to consider the phase noise caused by the vibration of ions leaving the equilibrium position and interacting with the laser, and the effect of this phase noise on the fidelity is not considered. Cryogenic ion trap systems, especially those with multiple DC electrodes, can fail to achieve optimal acoustic insulation and vibration attenuation, resulting in residual noise amplitudes close to the micrometer level.[61] To mitigate the impact of vibrational noise, Monroe et al. employed a synchronous shaking approach to minimize the phase discrepancy between the laser and the ions, as previously reported.[47] To mitigate and suppress time-dependent potential shaking, we developed a dynamic compensation technique that employs an inverse solution to ill-posed problems, enabling the real-time determination of compensation voltages by utilizing quadratic programming. The calculated voltages were used to stabilize the trapping potential. This approach effectively counteracted the translation induced by vibrational noise and facilitated the establishment of a stable superposed harmonic potential well. This technique has effectively demonstrated and suppressed the vibrational amplitude levels to within 20 nm of the potential minimum.

We propose an active suppression strategy for potential well shaking caused by vibrational noise. This strategy has the following advantages: it does not require the addition of a complex mechanical damping device and only needs to fully utilize the chip-based ion trap electrode that can be arbitrarily constructed to compensate for the potential, which can easily suppress the adverse effects of vibrational noise and is more flexible and convenient.

Our proposed dynamic compensation potential scheme is a theoretical solution aimed at suppressing residual vibrational noise. Future experimental efforts will assess the practical applications and effects of our theory within an experimental setup, with challenges focusing on the accuracy and bandwidth of noise measurements, accuracy of least-squares computed compensated electrode potentials, and bandwidth of feedback control. This approach is expected to significantly benefit applications in quantum error mitigation and the advancement of precision measurement devices, such as portable optical clocks, by suppressing vibrational noise.

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